Probability Rules & Conditional Probability

ANSWER KEY

Section 1: General Addition Rule & Mutually Exclusive Events

1 If $P(A) = 0.40$, $P(B) = 0.35$, and $P(A \cap B) = 0.15$, what is $P(A \cup B)$?

a. 0.60 ✓

b. 0.75

c. 0.90

d. 0.25

By the General Addition Rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.40 + 0.35 - 0.15 = 0.60$.

2 Two events E and F are mutually exclusive (disjoint). If $P(E) = 0.28$ and $P(F) = 0.45$, what is $P(E \cap F)$?

a. 0 ✓

b. 0.126

c. 0.73

d. 0.17

By definition, mutually exclusive events cannot happen simultaneously; therefore $P(E \cap F) = 0$.

3 Can two events with non-zero probabilities be both mutually exclusive and independent?

a. **No, if they are mutually exclusive, the occurrence of one prevents the other, meaning they are dependent** ✓

b. Yes, mutually exclusive and independent are synonyms in probability

c. Only if the sum of their probabilities equals 1

d. Yes, as long as their intersection is zero

If A and B are disjoint, $P(A \cap B) = 0$. For independence, $P(A \cap B) = P(A)P(B) > 0$. Since $0 \neq P(A)P(B)$, they cannot be both mutually exclusive and independent.

Section 2: Conditional Probability Formula

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4 If $P(A) = 0.50$ and $P(A \cap B) = 0.20$, what is the conditional probability $P(B \mid A)$?

a. 0.40 ✓

b. 0.10

c. 0.70

d. 0.25

By the conditional probability formula: $P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.20}{0.50} = 0.40$.

5 In a high school, 60% of students take math, and 18% take both math and art. Given that a randomly chosen student takes math, what is the probability they also take art?

a. 0.30 ✓

b. 0.18

c. 0.42

d. 0.108

$P(\text{Art} \mid \text{Math}) = \frac{P(\text{Art} \cap \text{Math})}{P(\text{Math})} = \frac{0.18}{0.60} = 0.30$.

6 Which statement mathematically proves that two events A and B are independent?

a. $P(A \mid B) = P(A)$ ✓

b. $P(A \cup B) = P(A) + P(B)$

c. $P(A \cap B) = 0$

d. $P(A) = P(B)$

Events A and B are independent if knowing that B occurred does not change the probability of A, meaning $P(A \mid B) = P(A)$ (or $P(A \cap B) = P(A)P(B)$).

Section 3: Multiplication Rule and Tree Diagrams

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7 A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn at random without replacement. What is the probability that both are red?

a. $\frac{5}{14}$ ✓

b. $\frac{25}{64}$

c. $\frac{5}{8}$

d. $\frac{10}{56}$

$P(R_1) = \frac{5}{8}$. Without replacement, $P(R_2 \mid R_1) = \frac{4}{7}$. $P(R_1 \cap R_2) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$.

8 If two events C and D are independent with $P(C) = 0.60$ and $P(D) = 0.70$, what is $P(C \cap D)$?

a. 0.42 ✓

b. 1.30

c. 0.10

d. 0.65

For independent events, $P(C \cap D) = P(C) \times P(D) = 0.60 \times 0.70 = 0.42$.

9 A factory has Machine A producing 60% of items (2% defective) and Machine B producing 40% (5% defective). What is the total probability that a randomly chosen item is defective?

a. 0.032 ✓

b. 0.070

c. 0.025

d. 0.012

By the Law of Total Probability:
$P(D) = P(A)P(D|A) + P(B)P(D|B) = (0.60)(0.02) + (0.40)(0.05) = 0.012 + 0.020 = 0.032$.

Section 4: Bayes' Theorem in Context

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10 Continuing the factory problem: if an item is found to be defective, what is the probability it came from Machine A?

a. $\frac{0.012}{0.032} = 0.375$ ✓

b. $\frac{0.60}{0.032}$

c. 0.60

d. 0.020

By Bayes' Theorem: $P(A \mid D) = \frac{P(A \cap D)}{P(D)} = \frac{0.012}{0.032} = \frac{3}{8} = 0.375$.

11 A diagnostic test has 99% sensitivity ($P(+|D) = 0.99$) and 95% specificity ($P(-|D^c) = 0.95$). If a rare disease affects 1% of the population ($P(D) = 0.01$), what is the probability that a person who tests positive actually has the disease?

a. ≈ 0.166 **(about** 16.6%**)** ✓

b. ≈ 0.990 (99%)

c. ≈ 0.950 (95%)

d. ≈ 0.500 (50%)

$P(+) = (0.01)(0.99) + (0.99)(0.05) = 0.0099 + 0.0495 = 0.0594$. Then $P(D \mid +) = \frac{0.0099}{0.0594} \approx 0.1667$.

12 What is $P(A^c \cup B^c)$ equivalent to by De Morgan's Laws?

a. $1 - P(A \cap B)$ ✓

b. $1 - P(A \cup B)$

c. $P(A) + P(B)$

d. $P(A \cap B)$

By De Morgan's Law, $A^c \cup B^c = (A \cap B)^c$. Therefore $P(A^c \cup B^c) = 1 - P(A \cap B)$.

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